

Getting Started with Git and GitHub pt. 1

Coffee, Cookie and Coding (C3)
Workshop supported by the Public
Health Data Science and Data
Equity team

Shelby Golden, M.S.

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Data Science and Data Equity



[xkcd Comic 1597 - Git](#) Downloaded April 26th, 2025.



Shelby Golden, M.S.

- Worked 7 years as a Molecular Biologist and Biochemist.
- Received a Masters in Applied Computational Mathematics from Johns Hopkins University in 2024.





Justin DeMayo

- System and Application Specialist for the Harvey Cushing/John Hay Whitney Medical Library.
- Offers introductory command line, git, and python courses.

Today's Learning Objectives

- 01 Understand the purpose and value of Git and GitHub in managing coding projects. (~ 5 minutes)
- 02 Learn how Git manages files for version control locally and distributes them through GitHub. (~ 15 minutes)
- 03 Set up and configure your local Git and GitHub accounts using either HTTPS or SSH Keys. (~ 40 minutes)



Our Choice Resources

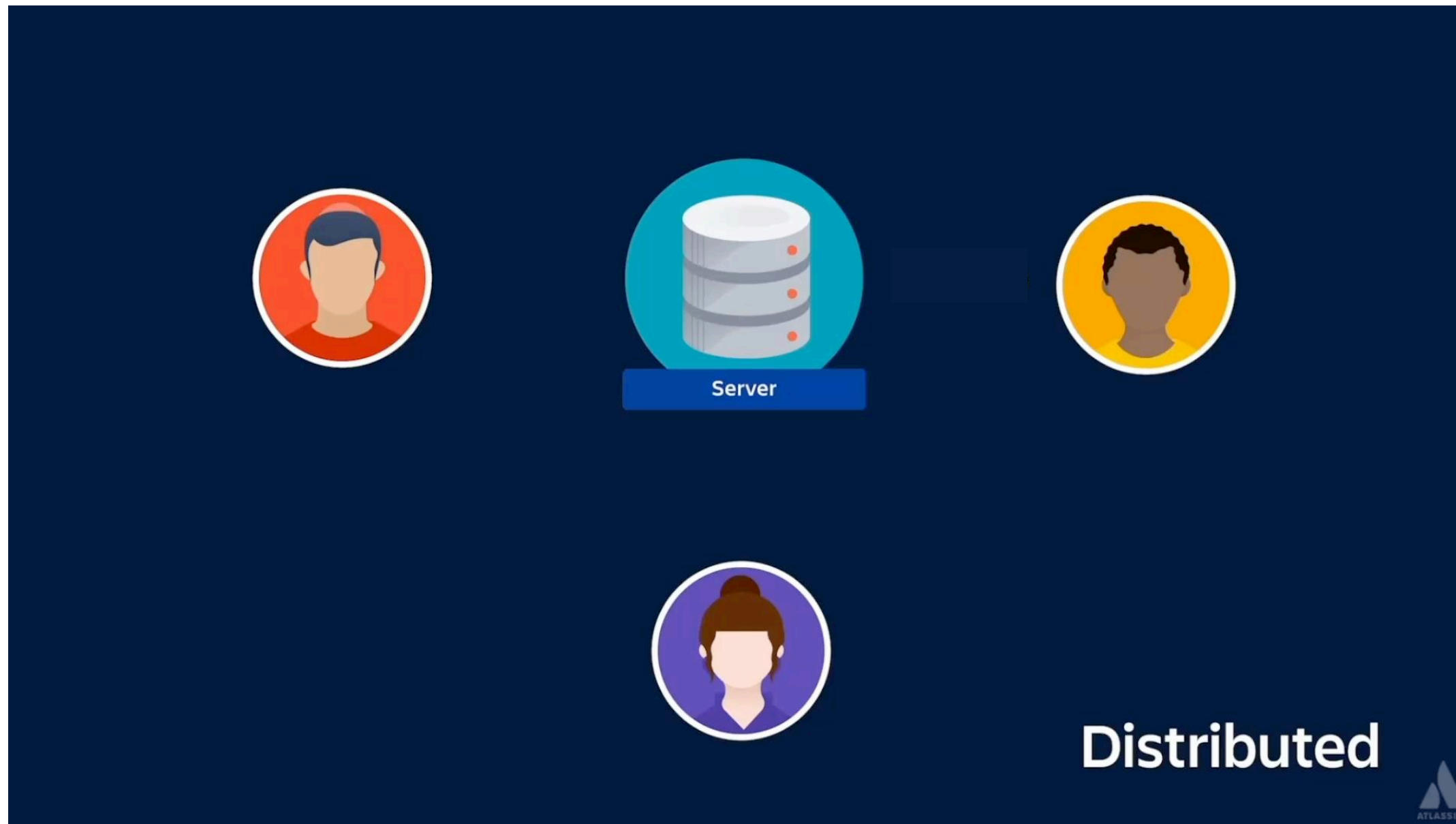
- Yale's Center for Research Computing workshop ["Version Control by Git"](#) by [Kaylea Nelson, Ph.D.](#)
- Yale's Harvey Cushing/John Hay Whitney Medical Library workshop ["Git & GitHub: An Introduction To Version Control"](#) by [Justin DeMayo](#)
- ["Getting Git Right"](#) by Atlassian
- [Git and GitHub tutorials](#) by W3Schools
- ["Introduction to GitHub"](#) by GitHub
- [Happy Git and GitHub for the useR](#) by [Professor Jenny Bryan](#) (and Yale alumni!) and [Jim Hester](#)

What is Git and GitHub?



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[What is Version Control](#) by Atlassian. Updated February 23rd, 2020.

[Git logo](#). Downloaded October 10th, 2024.

[GitHub logo](#). Downloaded October 10th, 2024.



System for project management by distributive version control (DVCS).



Developer platform for housing and managing projects and acts as the DVCS server.

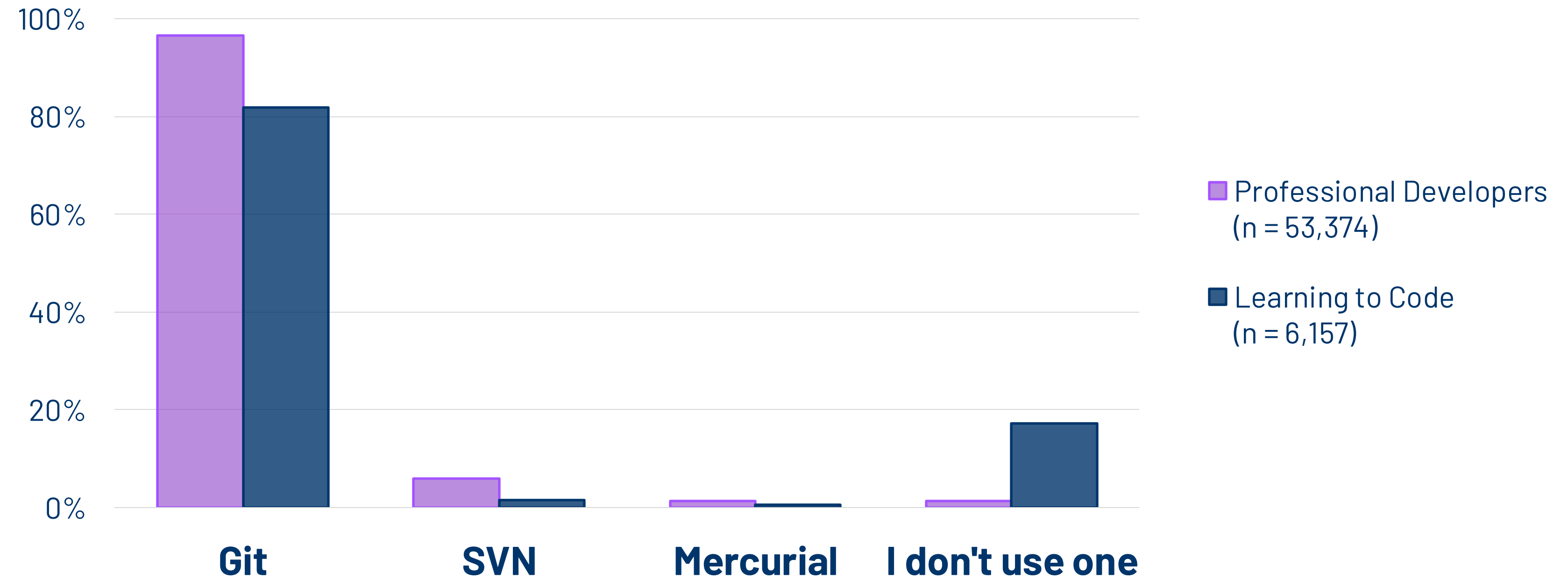


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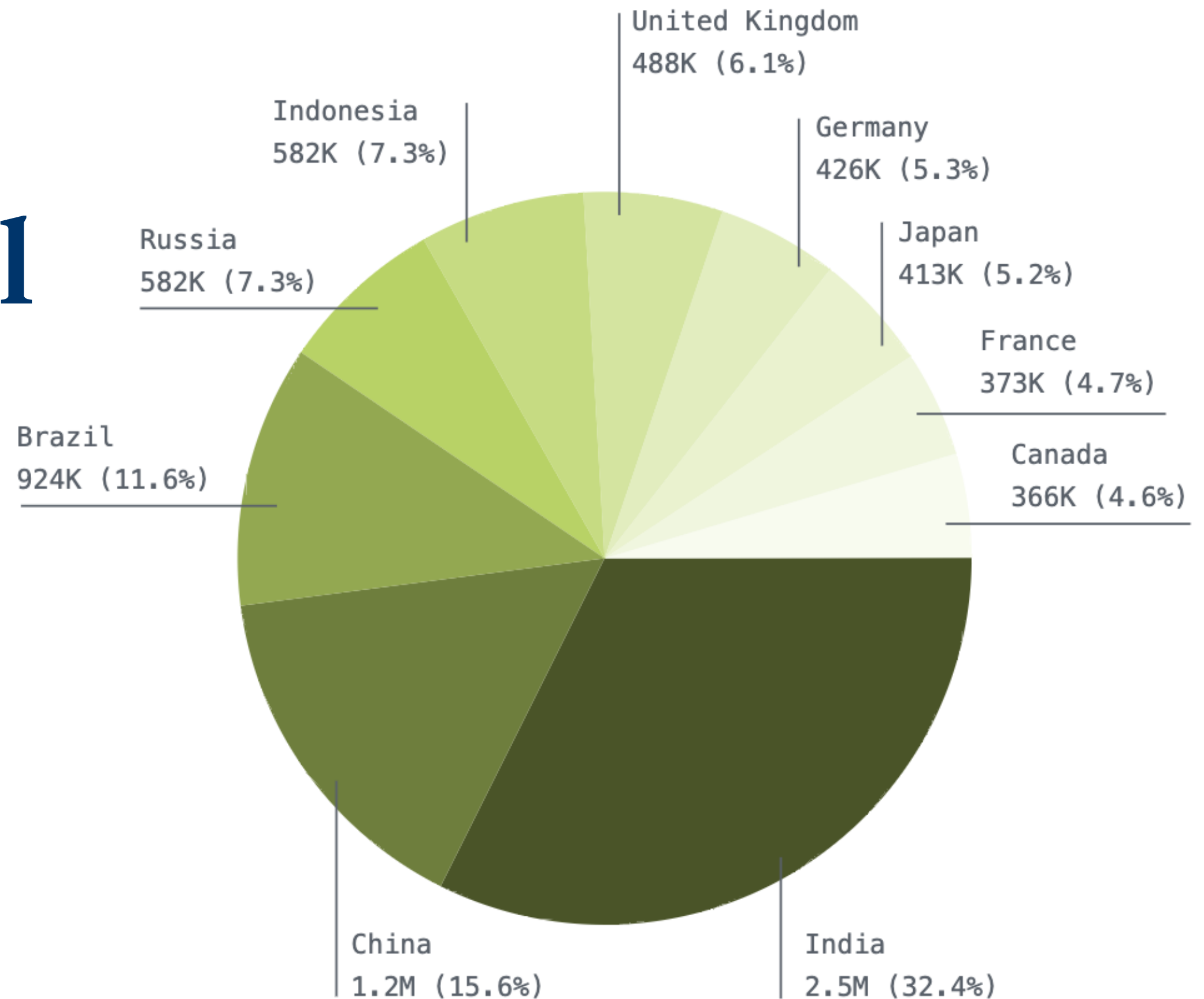
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Git is the most widely used VCS but is underutilized by those learning to code.

[2022 Developer Survey](#) by StackOverflow. Published 2022.



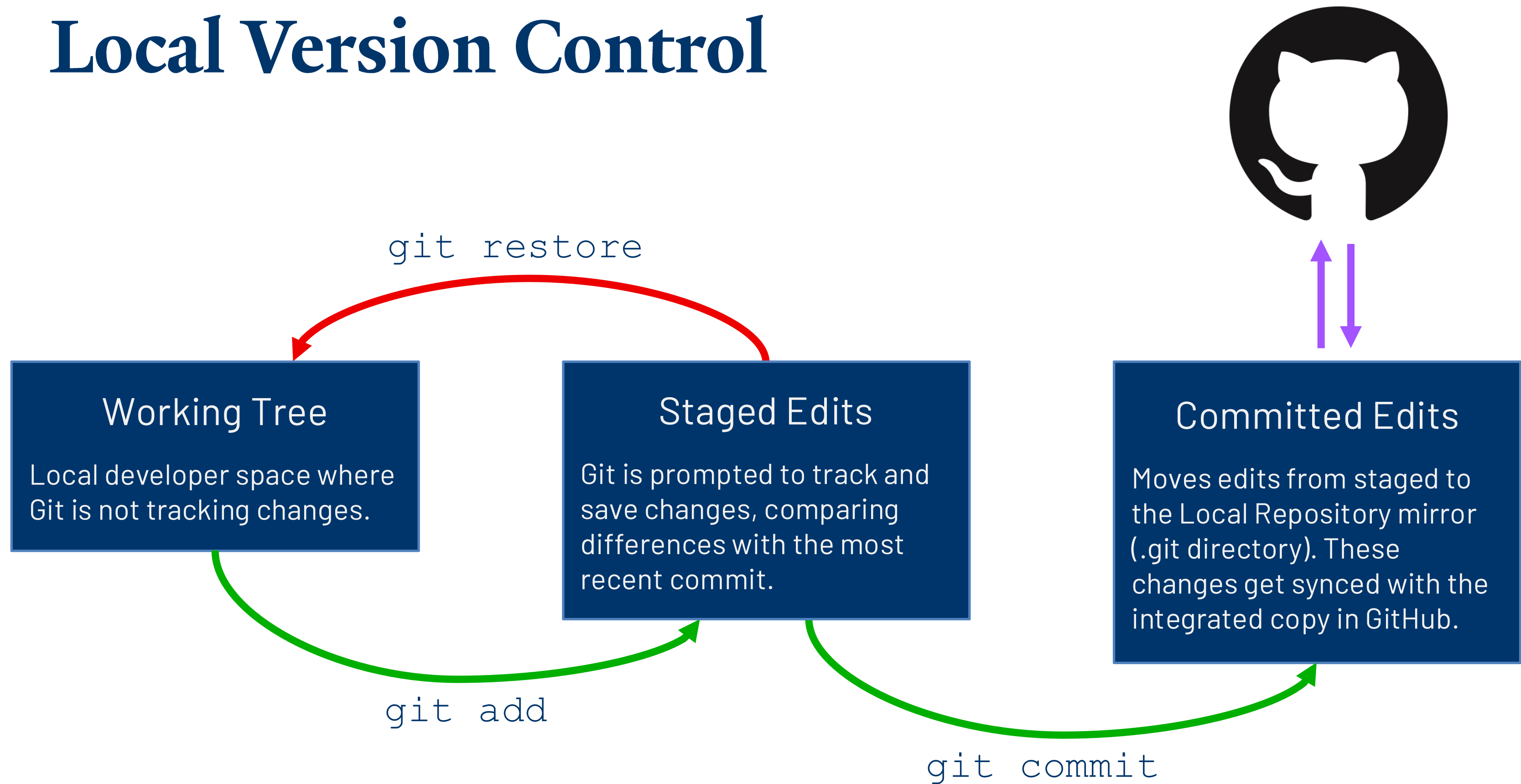
GitHub is an invaluable tool for coders worldwide.



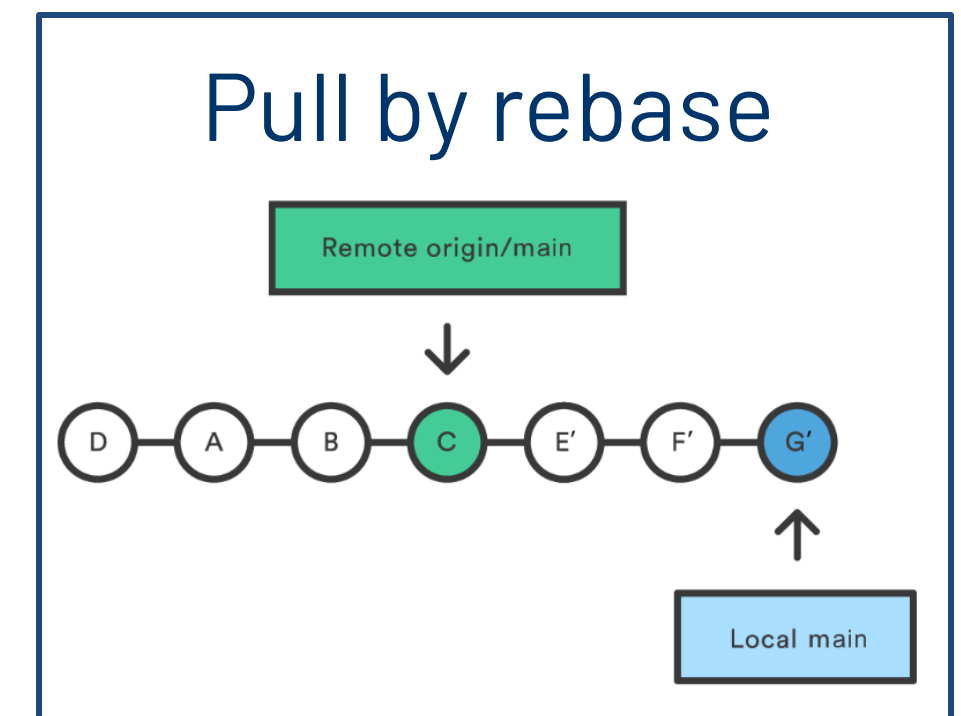
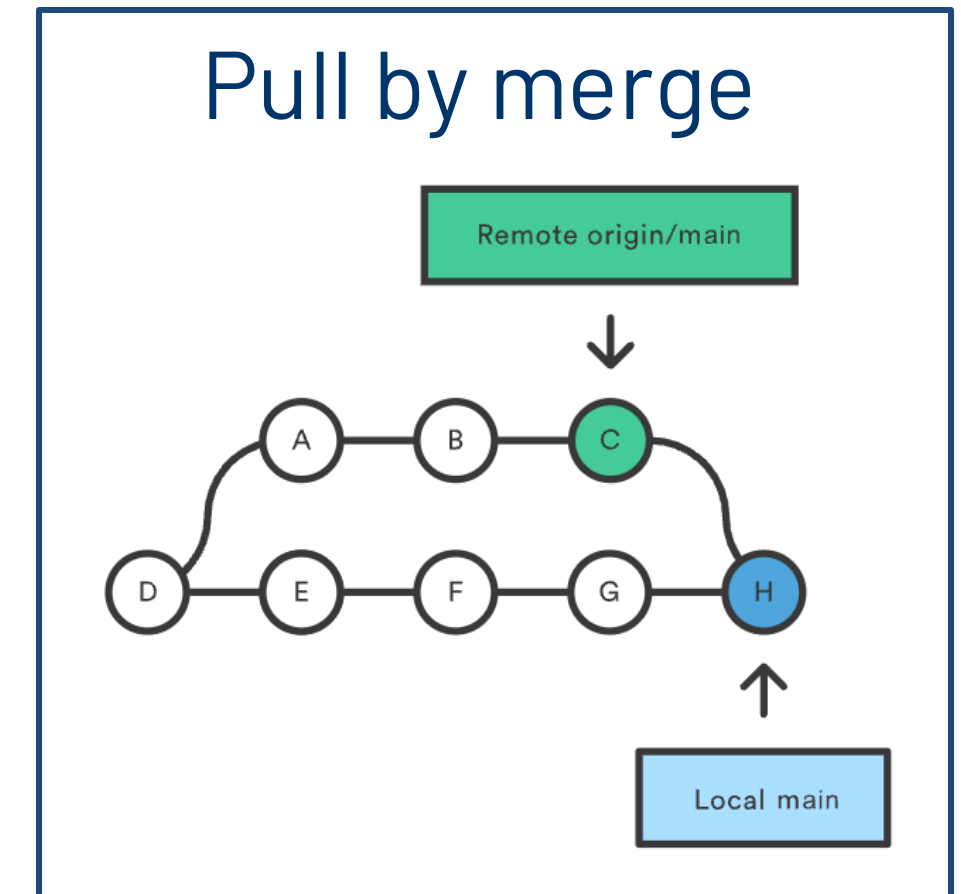
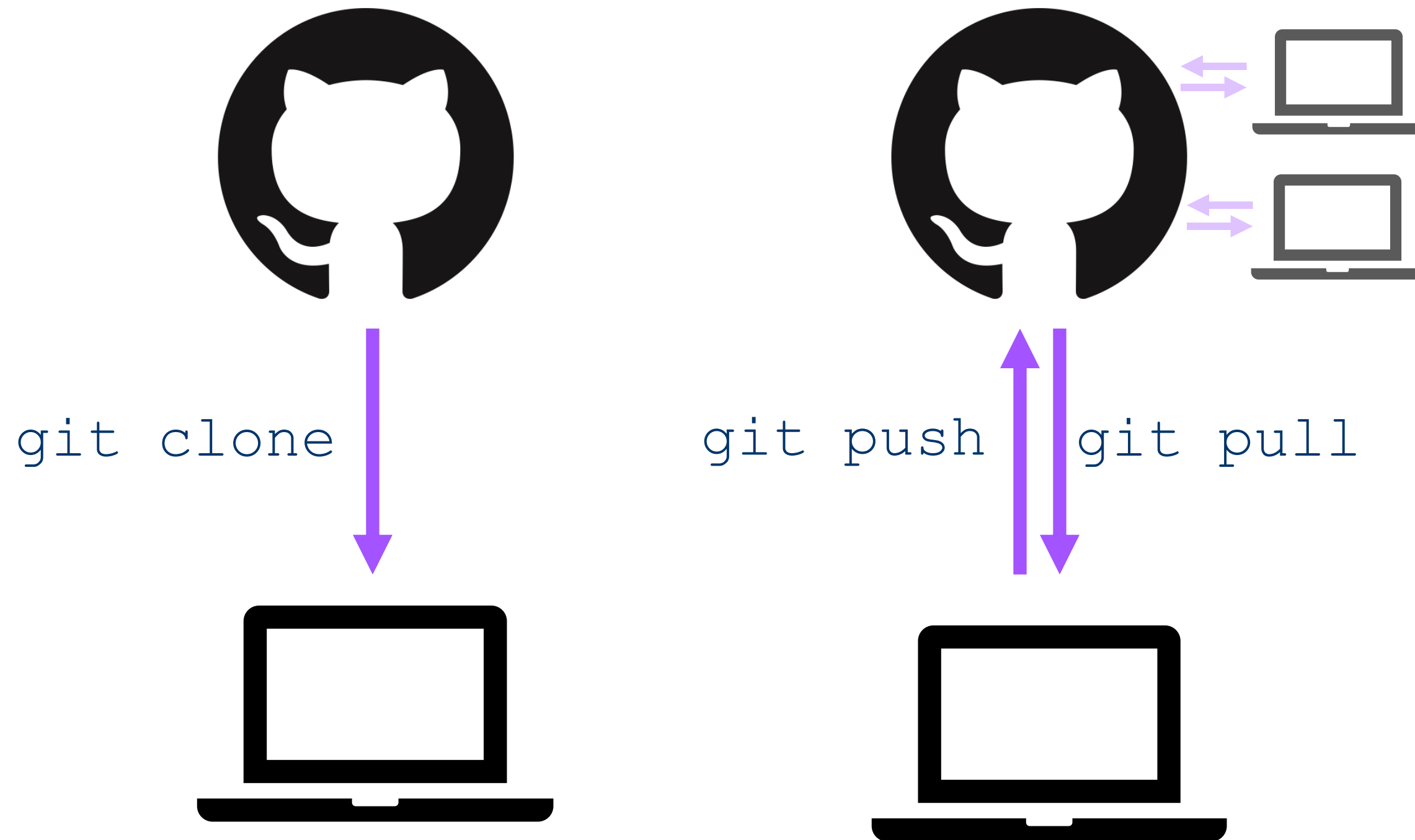
How do Git and GitHub relate to one another?



Local Version Control



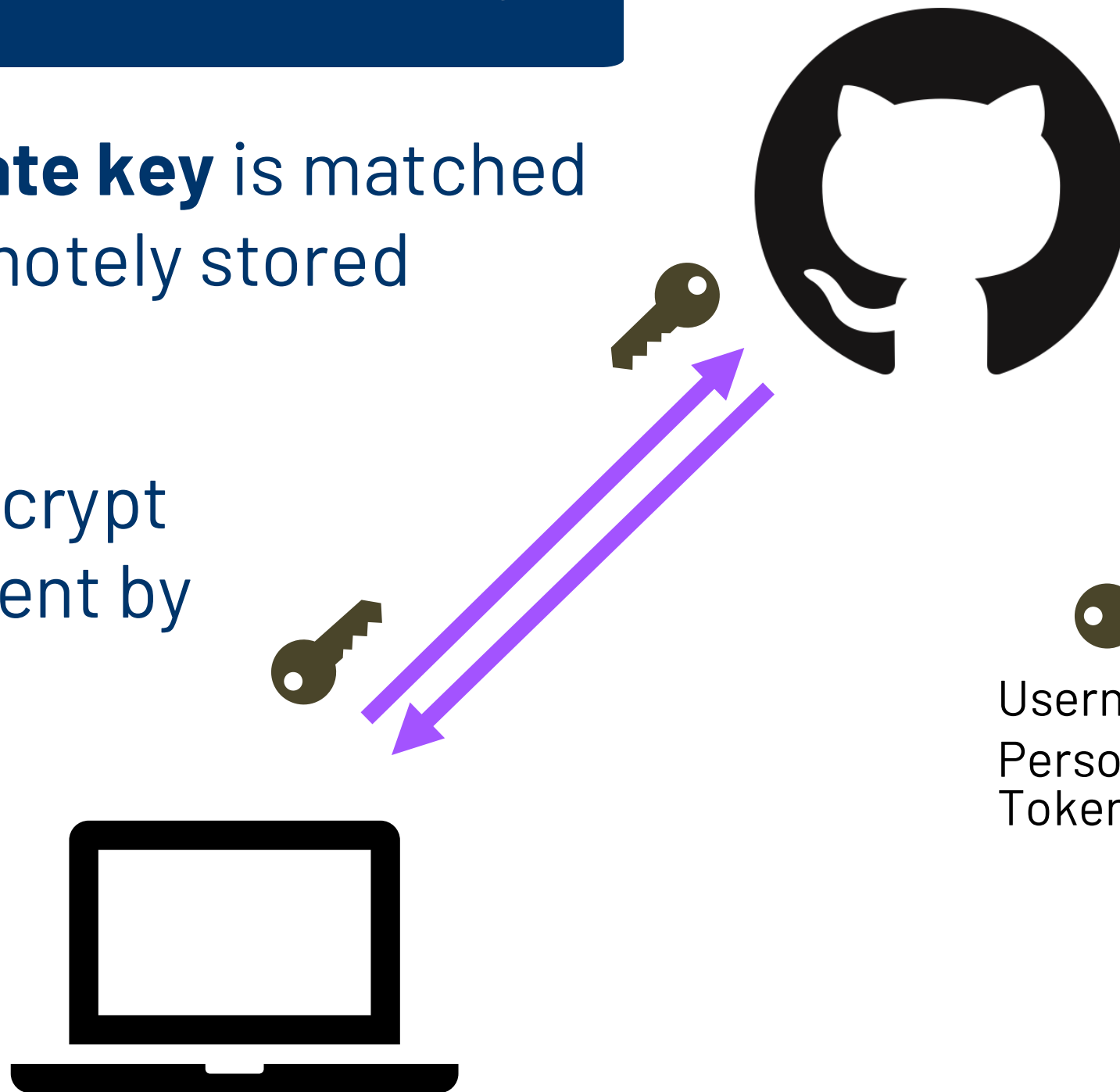
Remote Storage and Distribution



Secure Shell (SSH) Keys

A local **private key** is matched with the remotely stored **public key**.

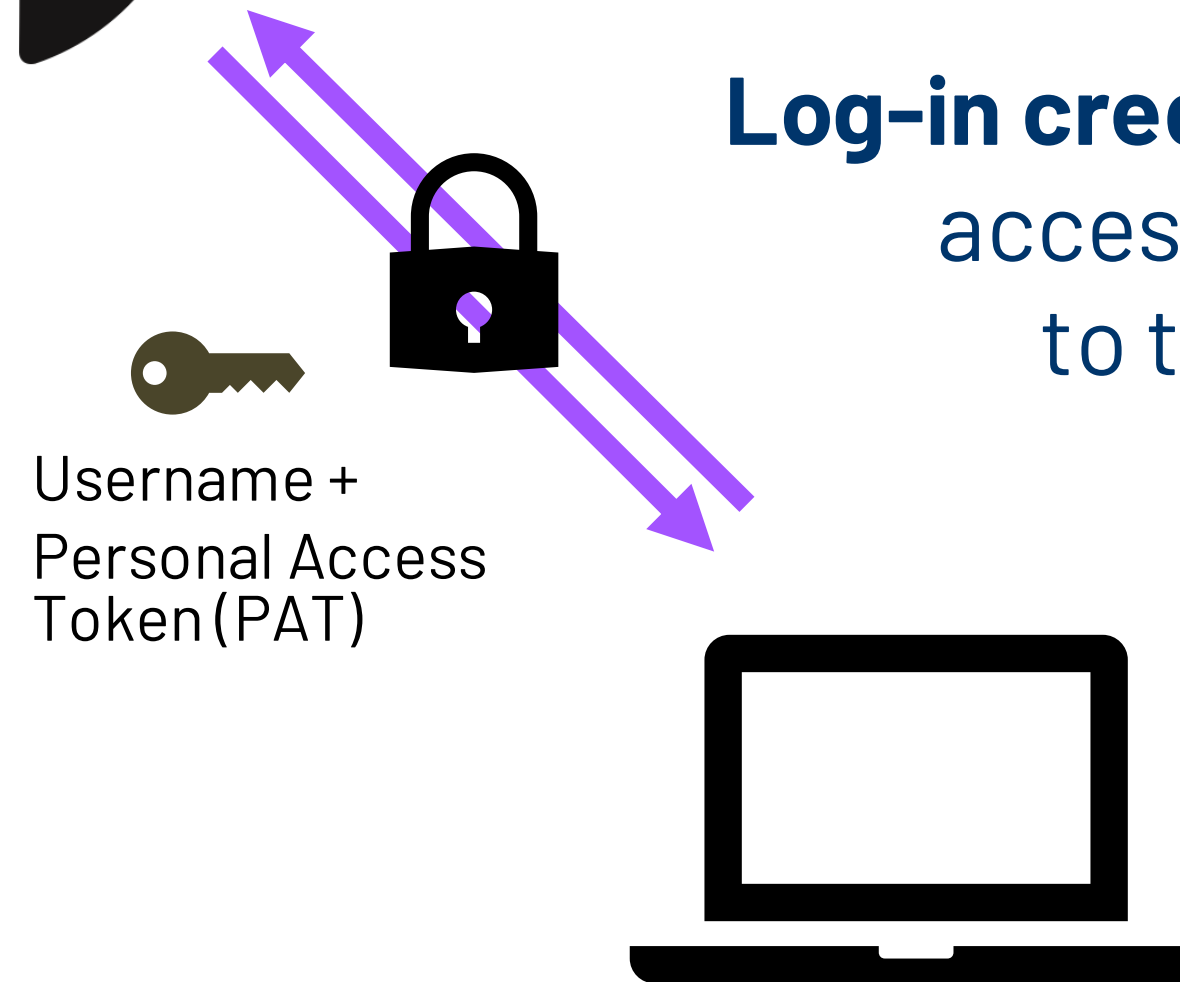
Keys can decrypt messages sent by their pair.



Hypertext Transfer Protocol Secure (HTTPS)

Information is securely transferred over the internet.

Log-in credentials verify access permissions to the repository.



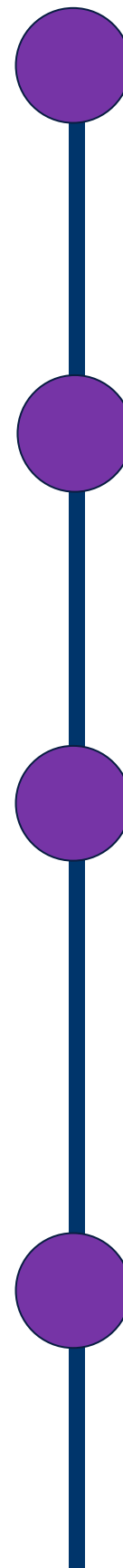
Additional Helpful Articles

- ["Cloning with SSH URL's" and "Troubleshooting SSH"](#) by GitHub
- ["Cloning with HTTPS URL's" and "Managing your PATs"](#) by GitHub
- ["About commit signature verification"](#) (sign [commits](#) and [tags](#)) by GitHub
- ["Managing deploy keys"](#) by GitHub
- ["Switching remote URLs from SSH to HTTPS"](#) and ["... SSH to HTTPS"](#) by GitHub
- ["Using SSH agent forwarding"](#) by GitHub
- ["Caching your GitHub credentials in Git"](#) for Windows by GitHub and ["Install the git-credential-osxkeychain helper"](#) for Macs by Atlassian

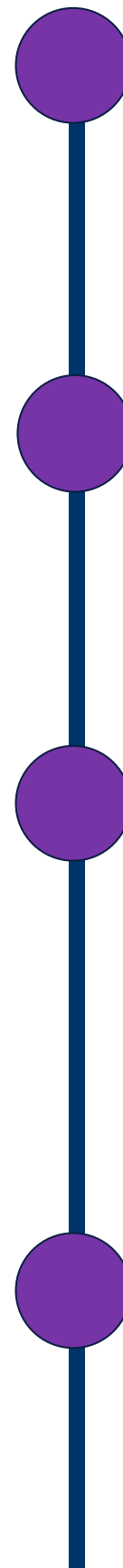
Credential Configuration: Let's Get Started!



Handout: Essential Setup

- 
- Install Git
 - Create a GitHub Account
 - HTTPS Personal Access Token (PAT)
 - SSH Key Pairs

Handout: Data Transfers

- 
- Install Git
 - Create a GitHub Account
 - HTTPS Personal Access Token (PAT)
 - SSH Key Pairs

Thank you!

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Appendix

Glossary

Command-Line Interface (CLI) A texted-base application that directly interacts with the computer's operating system, manages files, and can run programs. It typically lacks a graphical user interface (GUI).

Distributive Version Control System The project codebase is copied as a mirror to each contributor's local computer. Local changes get synched via patches sent "peer-to-peer" through the server.

git add Prompt Git to track changes made to specified files and transition them from the **Working Tree** to the **Staged Edits** domain. Git compares them to the `.git` directory.

Glossary

git branch Create or list different “versions” or “paths” of a project that you can work on separately.

git checkout Command to switch between the different branches.

git clone Copies an existing repository stored in a remote server to your own device, including all files, history, and branches.

git commit Promote the staged version of specified files to become the latest copy reflected in the `.git` directory. The **Committed Edits** version is synced to the remote server as a “peer-to-peer” patch.

Glossary

- git fetch** Downloads the remote branch contents into a separate local branch for review before integration. Part of `pull`.
- git merge** One method to reconcile different committing histories in divergent branches. Creates a new version integrating the head of the two branches in a three-way commit.
- git mv** Moves or renames a specified file, directory, or symlink that is already tracked by Git.
- git pull** Downloads a copy of the specified remote repository's branch and integrates it with the local copy. Combines the actions of `fetch` followed by `merge` or `rebase`.

Glossary

git push Upload your recent **Committed Edits** to a remote repository, synchronizing changes for others to see.

git rebase An alternative to `merge`. The branch commit histories are realigned so that the leading one defines the commit parent history of the following branch, thus rebasing its commits.

git reset Reverts content in the **Working Tree** or **Staged Edits** to their last committed state. Adding the `--staged` option will move files from **Staged Edits** back to the **Working Tree**, without overwriting any new changes there.

Glossary

git status Summarizes the state of files in the **Working Tree** and **Staged Edits**, comparing changes to the latest, committed version in the `.git` directory.

Graphical User Interface (GUI) An interface that allows users to interact with computers through visual elements like buttons and menus.

Hypertext Transfer Protocol Secure (HTTPS) Facilitates secure transmission of information over an encrypted internet connection. Decryption occurs when the correct username and Personal Access Token (PAT) are provided.

Glossary

Integrated Development Environment (IDE) A software application that combines tools for editing, building, testing, and debugging code into a single, user-friendly interface.

Mirror An exact copy of a project from a server, including all files, the version history, and branches.

Patch Snippets of code or data used to update existing software.

Peer-to-peer Participants in a network act as both client and server by trading resources and services with one another.

Glossary

Root directory The top-most directory in a branched hierarchy, containing all other files and directories. Your project's root directory holds all the relevant files and code used your project.

Server and Client Servers are computers or systems that provides resources (i.e. data or programs) to other computers, known as clients, over a network.

Shell A program used by the CLI to mediate communication between the user and computer by interpreting commands and outputs. Examples: Bash, PowerShell, etc.

Glossary

Secure Shell (SSH) Uses a pair of keys—a private key and a public key. Information is encrypted when sent between servers, and only the matching key pair can decrypt it upon receipt.

Version Control Manage, organize, and track different versions of files. Identify differences between versions and allows reverting to older versions.

Glossary

Helpful Cheat Sheets:

1. [Git Cheat Sheet](#) by Atlassian
2. [Vim Cheat Sheet](#) by [Richard Torruellas](#)
3. [Bash Shortcuts](#) by [Mohankumar Balasundaram](#)
4. [Command Line Cheat Sheet](#) by [Tobias Günther](#)

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